

# Deep Learning

## 1. Definition

Deep learning is a subfield of machine learning based on artificial neural networks with multiple layers (hence "deep"). Each layer learns increasingly abstract representations of the input data, removing the need for manual feature engineering.

## 2. Key Architectures

### 2.1 Feedforward Neural Networks (FNN)

Stacked fully-connected layers; the simplest deep architecture, used as a baseline and as the final classification head in most other networks.

### 2.2 Convolutional Neural Networks (CNN)

Use convolutional filters to exploit spatial locality. Dominant for image tasks; notable models include LeNet-5 (1998), AlexNet (2012), and ResNet (2015, introduced residual/skip connections).

### 2.3 Recurrent Neural Networks (RNN) and LSTM

Process sequential data by maintaining a hidden state across time steps. Long Short-Term Memory (LSTM) networks, introduced in 1997 by Hochreiter and Schmidhuber, use gating to mitigate the vanishing gradient problem.

### 2.4 Transformers

Introduced in the 2017 paper "Attention Is All You Need" (Vaswani et al.). Replace recurrence with self-attention, allowing full parallelization over sequence positions. Transformers underpin modern LLMs such as GPT and BERT.

## 3. Training Fundamentals

- Backpropagation computes gradients of the loss with respect to every weight via the chain rule.
- Gradient descent variants: SGD, Adam (2014), RMSProp update weights using these gradients.
- Regularization techniques: dropout (2014), batch normalization (2015), weight decay, and early stopping reduce overfitting.
- Learning rate schedules (cosine decay, warmup) stabilize training of large models.

## 4. Common Failure Modes

- Vanishing/exploding gradients in very deep or recurrent networks
- Overfitting when model capacity exceeds available training data
- Mode collapse in generative adversarial networks (GANs)
- Catastrophic forgetting during sequential/continual learning

## 5. Notable Frameworks

PyTorch (Meta, 2016) and TensorFlow (Google, 2015) are the two dominant deep learning frameworks; JAX (Google) is widely used for research requiring composable function transformations (grad, jit, vmap).